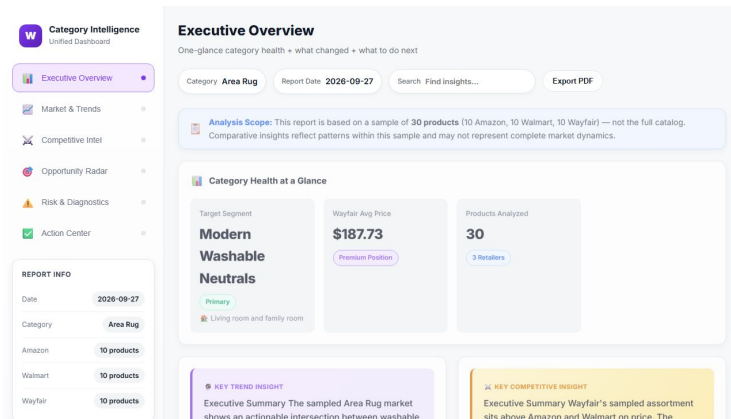


Wayfair Rugs Market Intelligence: AI Agent Demo

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Cohort: 2026



Executive Summary

Overview of Market Intelligence Agents

Objective

Give Wayfair's Rugs category team one workflow for category research and content decisions.

Solution

Built four connected n8n agents that turn public market signals into reviewable outputs.

Core Outputs

Trend report
Competitor report
Content strategy report
Integrated dashboard

Key Insights

30 comparable listings across Wayfair, Amazon, and Walmart

Wayfair sample average: \$187.73

Amazon: \$44.99; Walmart: \$30.45

Washable positioning is a visible competitive battleground

Future Improvements

Connect internal SKU, conversion, margin, return, and inventory data. Add scheduled monitoring with change alerts and human approval.

Agent 1: Visual Prompt Generator

From a product brief to a production-ready visual prompt

Objective

Convert a short design brief into a detailed, constrained prompt for rug moodboards and product visuals.

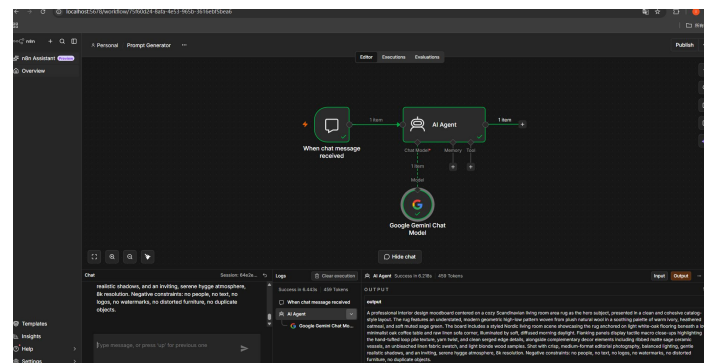
Input Prompt Example

"Cozy Scandinavian living room, area rug as hero subject"

Keep in Mind / Input Notes

The agent generates a visual prompt, not the final image. Users should specify style, room, product focus, lighting, and exclusions. Review the prompt before sending it to an image model.

Output Example: n8n workflow and generated prompt



 [View JSON Workflow](#)

 [View Submission Notes](#)

Agent 2: Trend Discovery

From public market signals to testable assortment hypotheses

Objective

Identify market, visual, and product-attribute signals for Wayfair's Area Rug category.

Input Prompt Example

"Area Rug / focus: washable"

Keep in Mind / Input Notes

Signals come from public sources and a controlled product sample. Treat findings as hypotheses. Validate with Wayfair sales, margin, returns, and review data.

Improvements / Expansion Opportunities

Add scheduled source refreshes, change detection, and category-level evaluation. Connect internal demand and return signals before scaling assortment recommendations.

 [View JSON Workflow](#)

 [View Trend Report](#)

Agent 3: Competitor Monitoring

Retailer listings into pricing and assortment evidence

Objective

Compare Wayfair with Amazon and Walmart using normalized product evidence.

Input Prompt Example

"Area Rug / focus: washable"

Keep in Mind / Input Notes

The sample contains 10 listings per retailer. It supports directional comparison, not a full market census. Margin, stock, delivery, and supplier terms remain outside scope.

Improvements / Expansion Opportunities

Track the same SKUs over time, add size-normalized pricing, and alert only on material assortment or price changes. Extend coverage after validating source reliability.

 [View JSON Workflow](#)

 [View Competitor Report](#)

Agent 4: AI Insights & Content

From market evidence to channel-ready content recommendations

Objective

Translate Project 2 and Project 3 evidence into Wayfair-ready messaging, campaigns, product copy, and quick wins.

Input Prompt Example

"Budget-conscious renters / Fall refresh / washable 5×7 rugs under \$200"

Keep in Mind / Input Notes

Outputs require brand, legal, and merchandising review. The workflow validates schema and keeps claims grounded in uploaded reports.

Improvements / Expansion Opportunities

Connect Wayfair's brand taxonomy and approval workflow. Evaluate claim accuracy, brand fit, and content performance by channel.

 [View JSON Workflow](#)

 [View Content Report](#)

Market Intelligence Dashboard

A single management view of trends, competitors, risks, and actions

Objective

Give category teams one review surface for the Project 2 trend report and Project 3 competitor report.

How This Works

Users upload both HTML reports. The agent parses controlled sections, validates required fields, fills the official dashboard template, and returns a downloadable HTML file.

Keep in Mind / Implementation Notes

The dashboard summarizes source evidence. It does not create new facts. Refreshes require new report uploads.

Improvements / Expansion Opportunities

Automate scheduled report ingestion, add version history, and connect internal KPIs so teams can compare external signals with business performance.

 [View JSON Workflow](#)

 [View Dashboard](#)

Reflections & Future Improvements

What I learned and where I would take the system next



Key Learnings

- I learned to separate evidence collection from AI interpretation.
- I improved agent reliability with deterministic parsing, schema validation, and clear fallbacks.
- I learned to frame market signals as hypotheses that require business validation.
- I practiced presenting technical workflows through business outcomes.



Future Improvements / Next Steps

- Connect SKU sales, margin, inventory, and return reasons.
- Schedule source refreshes and track changes over time.
- Add evaluation sets for extraction accuracy, recommendation grounding, and brand fit.
- Keep human approval before publishing or changing assortment.

Thank You

Questions and discussion

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